

Beyond Expected Goals: A Probabilistic Framework for Shot Occurrences in Soccer

Jonathan Pipping-Gamón¹ Tianshu Feng² Paul Sabin¹

¹Department of Statistics & Data Science, University of Pennsylvania

²Department of Computer & Information Science, University of Pennsylvania

January 31, 2026

What Does It Mean to “Deserve” a Win?

- **Post-match analysis:** We often hear “this team didn’t deserve that result.” But what does this really mean?

What Does It Mean to “Deserve” a Win?

- **Post-match analysis:** We often hear “this team didn’t deserve that result.” But what does this really mean?
- **Intuition:** Deserving refers to the difference in *goal-scoring danger* created or conceded by the two teams.

What Does It Mean to “Deserve” a Win?

- **Post-match analysis:** We often hear “this team didn’t deserve that result.” But what does this really mean?
- **Intuition:** Deserving refers to the difference in *goal-scoring danger* created or conceded by the two teams.
- **Mainstream approach:** Compare each team’s expected goals (xG)!

What Does It Mean to “Deserve” a Win?

- **Post-match analysis:** We often hear “this team didn’t deserve that result.” But what does this really mean?
- **Intuition:** Deserving refers to the difference in *goal-scoring danger* created or conceded by the two teams.
- **Mainstream approach:** Compare each team’s expected goals (xG)!
- **Our question:** Does xG really capture “deserving”?

Example 1: Comparing Chances

- **Match:** Manchester City vs. Real Madrid (Feb 19, 2025)

Example 1: Comparing Chances

- **Match:** Manchester City vs. Real Madrid (Feb 19, 2025)
- **Key question:** Which chance is more likely to produce a goal?

Example 2: Rebounded Shots

- **Match:** Orlando City vs. Philadelphia Union (Feb 22, 2025)

Example 2: Rebounded Shots

- **Match:** Orlando City vs. Philadelphia Union (Feb 22, 2025)
- **Key question:** How does xG measure the value of this sequence?

Example 2: Rebounded Shots

Time	Player	Shot Outcome (xG)
78:01	Brekalo	Shot Blocked (0.05)
78:02	Muriel	Shot Post (0.52)
78:04	Pasalic	Shot Post (0.68)
78:05	Pasalic	Shot Goal (0.38)

Table: Four shots in one possession totaling 1.63 xG, but only one goal is possible.

Example 3: Mbappé's Brilliance

- **Match:** Manchester City vs. Real Madrid (Feb 19, 2025)

Example 3: Mbappé's Brilliance

- **Match:** Manchester City vs. Real Madrid (Feb 19, 2025)
- **Key questions:** When is xG credited? What does it value?

What *Are* Expected Goals (xG)?

- **xG:** Probability that an *attempted shot* is scored, based on context

What *Are* Expected Goals (xG)?

- **xG:** Probability that an *attempted shot* is scored, based on context
 - Distance, angle, shot type, pressure, player positions, etc.

What *Are* Expected Goals (xG)?

- **xG:** Probability that an *attempted shot* is scored, based on context
 - Distance, angle, shot type, pressure, player positions, etc.
 - Often fitted using ML models (e.g., XGBoost) on historical shot data

What *Are* Expected Goals (xG)?

- **xG:** Probability that an *attempted shot* is scored, based on context
 - Distance, angle, shot type, pressure, player positions, etc.
 - Often fitted using ML models (e.g., XGBoost) on historical shot data
- **Purposes:**

What *Are* Expected Goals (xG)?

- **xG:** Probability that an *attempted shot* is scored, based on context
 - Distance, angle, shot type, pressure, player positions, etc.
 - Often fitted using ML models (e.g., XGBoost) on historical shot data
- **Purposes:**
 - Shot quality estimation

What Are Expected Goals (xG)?

- **xG:** Probability that an *attempted shot* is scored, based on context
 - Distance, angle, shot type, pressure, player positions, etc.
 - Often fitted using ML models (e.g., XGBoost) on historical shot data
- **Purposes:**
 - Shot quality estimation
 - Team performance summaries

What Are Expected Goals (xG)?

- **xG:** Probability that an *attempted shot* is scored, based on context
 - Distance, angle, shot type, pressure, player positions, etc.
 - Often fitted using ML models (e.g., XGBoost) on historical shot data
- **Purposes:**
 - Shot quality estimation
 - Team performance summaries
 - Finishing or goalkeeping “skill” via residuals (xGOE)

The Problem: xG Answers the Wrong Question

- As constructed, xG models are trained only on *observed* shots.

The Problem: xG Answers the Wrong Question

- As constructed, xG models are trained only on *observed* shots.
- So xG answers the following question:

$$\mathbb{P}(\text{goal} \mid \text{shot})$$

The Problem: xG Answers the Wrong Question

- As constructed, xG models are trained only on *observed* shots.
- So xG answers the following question:

$$\mathbb{P}(\text{goal} \mid \text{shot})$$

- But as we saw, many dangerous moments are *also* about:

$$\mathbb{P}(\text{shot})$$

The Problem: xG Answers the Wrong Question

- As constructed, xG models are trained only on *observed* shots.
- So xG answers the following question:

$$\mathbb{P}(\text{goal} \mid \text{shot})$$

- But as we saw, many dangerous moments are *also* about:

$$\mathbb{P}(\text{shot})$$

- What are the consequences of ignoring this?

The Problem: xG Answers the Wrong Question

- As constructed, xG models are trained only on *observed* shots.
- So xG answers the following question:

$$\mathbb{P}(\text{goal} \mid \text{shot})$$

- But as we saw, many dangerous moments are *also* about:

$$\mathbb{P}(\text{shot})$$

- What are the consequences of ignoring this?
 - No-shot danger gets 0 credit (Example 1)

The Problem: xG Answers the Wrong Question

- As constructed, xG models are trained only on *observed* shots.
- So xG answers the following question:

$$\mathbb{P}(\text{goal} \mid \text{shot})$$

- But as we saw, many dangerous moments are *also* about:

$$\mathbb{P}(\text{shot})$$

- What are the consequences of ignoring this?
 - No-shot danger gets 0 credit (Example 1)
 - Rebound sequences inflate summed xG (Example 2)

The Problem: xG Answers the Wrong Question

- As constructed, xG models are trained only on *observed* shots.
- So xG answers the following question:

$$\mathbb{P}(\text{goal} \mid \text{shot})$$

- But as we saw, many dangerous moments are *also* about:

$$\mathbb{P}(\text{shot})$$

- What are the consequences of ignoring this?
 - No-shot danger gets 0 credit (Example 1)
 - Rebound sequences inflate summed xG (Example 2)
 - The skill of shot *creation* is under-credited (Example 3)

Modeling the *True* Goal-Scoring Process

- Scoring can be thought of as a two-step process:

Modeling the *True* Goal-Scoring Process

- Scoring can be thought of as a two-step process:
 - 1 **Will a shot happen?** (creating a chance)

Modeling the *True* Goal-Scoring Process

- Scoring can be thought of as a two-step process:
 - ➊ **Will a shot happen?** (creating a chance)
 - ➋ **If it happens, how likely is it to be a goal?** (shot quality)

Modeling the *True* Goal-Scoring Process

- Scoring can be thought of as a two-step process:
 - ➊ **Will a shot happen?** (creating a chance)
 - ➋ **If it happens, how likely is it to be a goal?** (shot quality)
- **We should model both.**

Modeling the *True* Goal-Scoring Process

- Scoring can be thought of as a two-step process:
 - 1 **Will a shot happen?** (creating a chance)
 - 2 **If it happens, how likely is it to be a goal?** (shot quality)
- **We should model both.**
 - xS : the probability a shot occurs in the next second

Modeling the *True* Goal-Scoring Process

- Scoring can be thought of as a two-step process:
 - 1 **Will a shot happen?** (creating a chance)
 - 2 **If it happens, how likely is it to be a goal?** (shot quality)
- **We should model both.**
 - xS : the probability a shot occurs in the next second
 - xG : the probability of a goal given a shot occurs

Modeling the *True* Goal-Scoring Process

- Scoring can be thought of as a two-step process:
 - ➊ **Will a shot happen?** (creating a chance)
 - ➋ **If it happens, how likely is it to be a goal?** (shot quality)
- **We should model both.**
 - xS : the probability a shot occurs in the next second
 - xG : the probability of a goal given a shot occurs
- Putting them together produces **$xG+$** : the probability of scoring in the next second.

Introducing Our New Metric: xG+

- At each time point t , define:

$$\begin{aligned} \text{xG+}_t &= \mathbb{P}_t(\text{goal scored}) \\ &= \mathbb{P}_t(\text{goal} \mid \text{shot}) \cdot \mathbb{P}_t(\text{shot}) \\ &= \text{xG}_t \cdot \text{xS}_t \end{aligned}$$

Introducing Our New Metric: xG+

- At each time point t , define:

$$\begin{aligned} \text{xG+}_t &= \mathbb{P}_t(\text{goal scored}) \\ &= \mathbb{P}_t(\text{goal} \mid \text{shot}) \cdot \mathbb{P}_t(\text{shot}) \\ &= \text{xG}_t \cdot \text{xS}_t \end{aligned}$$

- Possession-level aggregation (over n time points):**

$$\text{xG+}_{\text{poss}} = 1 - \prod_{t=1}^n (1 - \text{xG+}_t)$$

Introducing Our New Metric: xG+

- At each time point t , define:

$$\begin{aligned} \text{xG+}_t &= \mathbb{P}_t(\text{goal scored}) \\ &= \mathbb{P}_t(\text{goal} \mid \text{shot}) \cdot \mathbb{P}_t(\text{shot}) \\ &= \text{xG}_t \cdot \text{xS}_t \end{aligned}$$

- Possession-level aggregation (over n time points):**

$$\text{xG+}_{\text{poss}} = 1 - \prod_{t=1}^n (1 - \text{xG+}_t)$$

- What this gives us: the probability of scoring (at most once) in the possession.

Introducing Our New Metric: xG+

- At each time point t , define:

$$\begin{aligned} \text{xG+}_t &= \mathbb{P}_t(\text{goal scored}) \\ &= \mathbb{P}_t(\text{goal} \mid \text{shot}) \cdot \mathbb{P}_t(\text{shot}) \\ &= \text{xG}_t \cdot \text{xS}_t \end{aligned}$$

- Possession-level aggregation (over n time points):**

$$\text{xG+}_{\text{poss}} = 1 - \prod_{t=1}^n (1 - \text{xG+}_t)$$

- What this gives us: the probability of scoring (at most once) in the possession.
- Respects the fundamental limit of at most one goal per possession.

- **Source:** Gradient Sports tracking and event data (EPL 2022–2025)

- **Source:** Gradient Sports tracking and event data (EPL 2022–2025)
- **Resolution:** 30 frames per second

- **Source:** Gradient Sports tracking and event data (EPL 2022–2025)
- **Resolution:** 30 frames per second
- **Inputs:**

- **Source:** Gradient Sports tracking and event data (EPL 2022–2025)
- **Resolution:** 30 frames per second
- **Inputs:**
 - Player positions (x,y), ball position (x,y,z)

- **Source:** Gradient Sports tracking and event data (EPL 2022–2025)
- **Resolution:** 30 frames per second
- **Inputs:**
 - Player positions (x,y), ball position (x,y,z)
 - Possession indicators, shot events and outcomes

- **Source:** Gradient Sports tracking and event data (EPL 2022–2025)
- **Resolution:** 30 frames per second
- **Inputs:**
 - Player positions (x,y), ball position (x,y,z)
 - Possession indicators, shot events and outcomes
 - Player/team IDs

- Keep frames where the ball is in play and a team has *clear possession* in the attacking third

- Keep frames where the ball is in play and a team has *clear possession* in the attacking third
- Define attacking sequences:

- Keep frames where the ball is in play and a team has *clear possession* in the attacking third
- Define attacking sequences:
 - Start: possession gained in attacking third

- Keep frames where the ball is in play and a team has *clear possession* in the attacking third
- Define attacking sequences:
 - Start: possession gained in attacking third
 - End: the defense regains possession or the ball exits the attacking third

- Keep frames where the ball is in play and a team has *clear possession* in the attacking third
- Define attacking sequences:
 - Start: possession gained in attacking third
 - End: the defense regains possession or the ball exits the attacking third
- Interpolate ball positions to fill in missing frames

- **Ball details:** distance to goal, angle, height, speed

- **Ball details:** distance to goal, angle, height, speed
- **Local pressure:** nearest attackers and defenders relative to ball

- **Ball details:** distance to goal, angle, height, speed
- **Local pressure:** nearest attackers and defenders relative to ball
- **GK context:** keeper location relative to goal

- **Ball details:** distance to goal, angle, height, speed
- **Local pressure:** nearest attackers and defenders relative to ball
- **GK context:** keeper location relative to goal
- **Goal openness (openGoal):**

- **Ball details:** distance to goal, angle, height, speed
- **Local pressure:** nearest attackers and defenders relative to ball
- **GK context:** keeper location relative to goal
- **Goal openness (openGoal):**
 - Model defenders as circles (diameter 0.75 m)

- **Ball details:** distance to goal, angle, height, speed
- **Local pressure:** nearest attackers and defenders relative to ball
- **GK context:** keeper location relative to goal
- **Goal openness (openGoal):**
 - Model defenders as circles (diameter 0.75 m)
 - Compute fraction of goalmouth left unobstructed by defenders

- We train two supervised models using the same feature set:

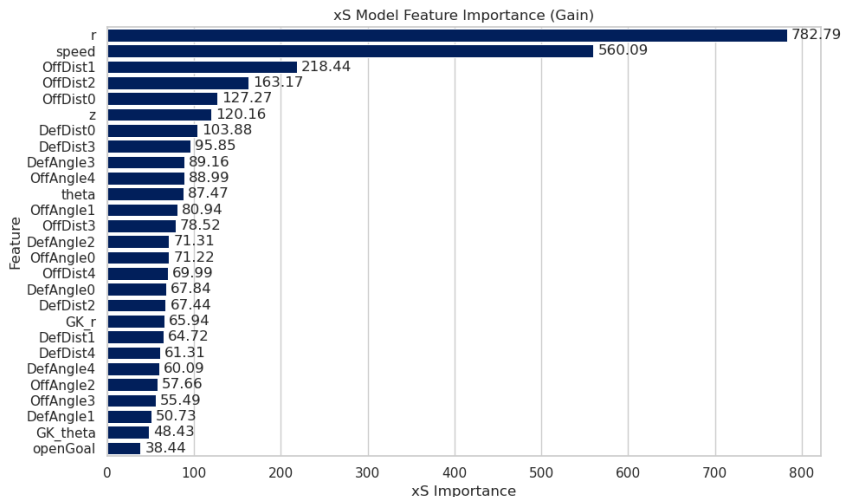
- We train two supervised models using the same feature set:
 - **xS**: $\mathbb{P}(\text{shot in the next second})$

- We train two supervised models using the same feature set:
 - **xS**: $\mathbb{P}(\text{shot in the next second})$
 - **xG**: $\mathbb{P}(\text{goal} \mid \text{shot})$

- We train two supervised models using the same feature set:
 - **xS**: $\mathbb{P}(\text{shot in the next second})$
 - **xG**: $\mathbb{P}(\text{goal} \mid \text{shot})$
- Fit using 5-fold cross-validated XGBoost

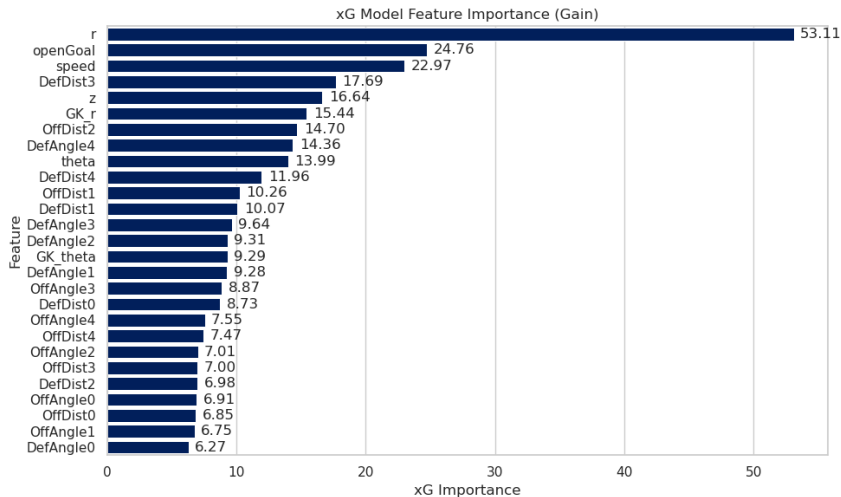
- We train two supervised models using the same feature set:
 - **xS**: $\mathbb{P}(\text{shot in the next second})$
 - **xG**: $\mathbb{P}(\text{goal} \mid \text{shot})$
- Fit using 5-fold cross-validated XGBoost
- Evaluated with log loss (lower is better)

What Drives Shot Creation (xS)?



- Distance and ball speed dominate: proximity and tempo create chances.

What Drives Shot Quality (xG)?



- Distance, openGoal, and ball speed primarily determine shot quality.

Does $xG+$ Predict Goals Better than xG ?

- **Idea:** If a metric measures “goal-scoring danger,” it should predict goals well!

Does $xG+$ Predict Goals Better than xG ?

- **Idea:** If a metric measures “goal-scoring danger,” it should predict goals well!
- **Data:** 3 EPL seasons, 114 matchdays.

Does $xG+$ Predict Goals Better than xG ?

- **Idea:** If a metric measures “goal-scoring danger,” it should predict goals well!
- **Data:** 3 EPL seasons, 114 matchdays.
- **Out-of-sample test:**

Does xG+ Predict Goals Better than xG?

- **Idea:** If a metric measures “goal-scoring danger,” it should predict goals well!
- **Data:** 3 EPL seasons, 114 matchdays.
- **Out-of-sample test:**
 - 1 Hold out one matchday at a time.

Does xG+ Predict Goals Better than xG?

- **Idea:** If a metric measures “goal-scoring danger,” it should predict goals well!
- **Data:** 3 EPL seasons, 114 matchdays.
- **Out-of-sample test:**
 - 1 Hold out one matchday at a time.
 - 2 Train our models on the other 113 matchdays.

Does xG+ Predict Goals Better than xG?

- **Idea:** If a metric measures “goal-scoring danger,” it should predict goals well!
- **Data:** 3 EPL seasons, 114 matchdays.
- **Out-of-sample test:**
 - 1 Hold out one matchday at a time.
 - 2 Train our models on the other 113 matchdays.
 - 3 Predict goals on the held-out matchday.

Does $xG+$ Predict Goals Better than xG ?

- **Idea:** If a metric measures “goal-scoring danger,” it should predict goals well!
- **Data:** 3 EPL seasons, 114 matchdays.
- **Out-of-sample test:**
 - 1 Hold out one matchday at a time.
 - 2 Train our models on the other 113 matchdays.
 - 3 Predict goals on the held-out matchday.
- **Compare:** xG vs xS vs $xG+$ (with two possession aggregation rules).

Moment-level metrics:

- **xS**: Probability of a shot in the next second

Possession aggregation:

Moment-level metrics:

- **xS:** Probability of a shot in the next second
- **xG:** Probability of scoring given a shot

Possession aggregation:

Moment-level metrics:

- **xS**: Probability of a shot in the next second
- **xG**: Probability of scoring given a shot
- **xG+**: $xS \times xG$ (probability of scoring in the next second)

Possession aggregation:

Moment-level metrics:

- **xS:** Probability of a shot in the next second
- **xG:** Probability of scoring given a shot
- **xG+:** $xS \times xG$ (probability of scoring in the next second)

Possession aggregation:

- **At-least-one-per-possession:** $1 - \prod(1 - p)$

Moment-level metrics:

- **xS:** Probability of a shot in the next second
- **xG:** Probability of scoring given a shot
- **xG+:** $xS \times xG$ (probability of scoring in the next second)

Possession aggregation:

- **At-least-one-per-possession:** $1 - \prod(1 - p)$
- **Max-per-possession:** $\max(p)$

Moment-level metrics:

- **xS:** Probability of a shot in the next second
- **xG:** Probability of scoring given a shot
- **xG+:** $xS \times xG$ (probability of scoring in the next second)

Possession aggregation:

- **At-least-one-per-possession:** $1 - \prod(1 - p)$
- **Max-per-possession:** $\max(p)$
- **Baseline:** sum of shot-by-shot xG

Team-Level Results: $xG+ > xS > xG$

Table: Mean Squared Error (MSE) by Metric and Aggregation Method

Aggregation Method	$xG+$	xS	xG
At-least-one-per-possession	2.84	2.90	2.94
Max-per-possession	2.84	2.87	2.91
Sum-of-shots			2.90

- **Takeaway:** Both $xG+$ and xS dominate xG , with $xG+$ as the best overall.

What About Player Ability?

- **Question:** Do we learn more about players from *shot-taking* or *goal-scoring*?

What About Player Ability?

- **Question:** Do we learn more about players from *shot-taking* or *goal-scoring*?
- **Approach:**

What About Player Ability?

- **Question:** Do we learn more about players from *shot-taking* or *goal-scoring*?
- **Approach:**
 - ➊ Aggregate each metric by player and season (xG , xS , $xG+$)

What About Player Ability?

- **Question:** Do we learn more about players from *shot-taking* or *goal-scoring*?
- **Approach:**
 - 1 Aggregate each metric by player and season (xG , xS , $xG+$)
 - 2 Compare to observed goals and shots ($xGOE$, $xSOE$, $xG+OE$)

What About Player Ability?

- **Question:** Do we learn more about players from *shot-taking* or *goal-scoring*?
- **Approach:**
 - 1 Aggregate each metric by player and season (xG , xS , $xG+$)
 - 2 Compare to observed goals and shots ($xGOE$, $xSOE$, $xG+OE$)
 - 3 Divide by matches played to get per-game metrics

What About Player Ability?

- **Question:** Do we learn more about players from *shot-taking* or *goal-scoring*?
- **Approach:**
 - 1 Aggregate each metric by player and season (xG , xS , $xG+$)
 - 2 Compare to observed goals and shots ($xGOE$, $xSOE$, $xG+OE$)
 - 3 Divide by matches played to get per-game metrics
 - 4 Evaluate year-over-year stability (repeatability)

Key Finding: Finishing Is Not Stable; Shot-Taking Is

Table: Year-to-Year Correlation (Stability) of Each Metric (Per Game)

xGOE	xSOE	xG+OE
0.12	0.63	0.35

- **Interpretation:**

Key Finding: Finishing Is Not Stable; Shot-Taking Is

Table: Year-to-Year Correlation (Stability) of Each Metric (Per Game)

xGOE	xSOE	xG+OE
0.12	0.63	0.35

- **Interpretation:**

- **Finishing is very noisy:** xGOE is weakly correlated year-to-year (0.12).

Key Finding: Finishing Is Not Stable; Shot-Taking Is

Table: Year-to-Year Correlation (Stability) of Each Metric (Per Game)

xGOE	xSOE	xG+OE
0.12	0.63	0.35

- **Interpretation:**

- **Finishing is very noisy:** xGOE is weakly correlated year-to-year (0.12).
- **Shot-taking is repeatable:** xSOE is strongly correlated (0.63).

Key Finding: Finishing Is Not Stable; Shot-Taking Is

Table: Year-to-Year Correlation (Stability) of Each Metric (Per Game)

xGOE	xSOE	xG+OE
0.12	0.63	0.35

- **Interpretation:**

- **Finishing is very noisy:** xGOE is weakly correlated year-to-year (0.12).
- **Shot-taking is repeatable:** xSOE is strongly correlated (0.63).
- **Takeaway:** Top players are consistently great because they *create more shots than expected*, not because they reliably “finish hot.”

Top Overperforming Players

xG+ Over Expected (Per Game)

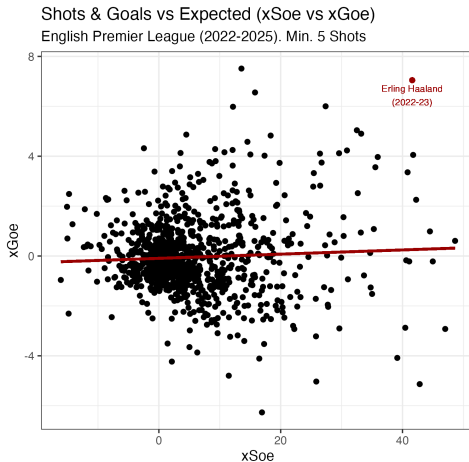
Season	Player	xG+ OE	Goals	Games	Chances
2022-23	Erling Haaland	0.52	36	35	444
2023-24	Erling Haaland	0.44	30	31	513
2024-25	Omar Marmoush	0.41	10	16	293
2024-25	Yoane Wissa	0.35	22	35	429
2024-25	Mohamed Salah	0.33	27	38	1042
2024-25	Chris Wood	0.31	23	36	330
2023-24	Cole Palmer	0.30	19	34	693
2024-25	Alexander Isak	0.30	25	34	558
2023-24	Alexander Isak	0.30	19	28	351
2022-23	Harry Kane	0.29	25	38	589

xG Over Expected (Per Game)

Season	Player	xG OE	Goals	Games	Chances
2024-25	Omar Marmoush	0.41	10	16	293
2024-25	Chris Wood	0.31	23	36	330
2024-25	Michael Keane	0.21	3	10	19
2022-23	Erling Haaland	0.52	36	35	444
2022-23	Roberto Firmino	0.25	10	21	253
2022-23	Martin Ødegaard	0.22	15	37	942
2023-24	Heung-min Son	0.28	20	35	751
2022-23	Matias Viña	0.26	3	10	66
2022-23	Alexander Isak	0.24	11	22	346
2023-24	Taiwo Awoniyi	0.26	7	19	125

Note: Minimum 10 Games with a Chance

Example: Haaland's Record Season



- **Key insight:** Required both taking more shots than expected and favorable finishing variance

Conclusions

- **Best metric for team prediction: $xG+$**

Conclusions

- **Best metric for team prediction:** $xG+$
- **Best aggregation method:** at-least-one-per-possession

Conclusions

- **Best metric for team prediction:** $xG+$
- **Best aggregation method:** at-least-one-per-possession
- **Key insight:** Shot creation is more stable than finishing

Conclusions

- **Best metric for team prediction:** $xG+$
- **Best aggregation method:** at-least-one-per-possession
- **Key insight:** Shot creation is more stable than finishing
- **Why this matters:** match analysis, player evaluation, recruitment, and forecasting

Conclusions

- **Best metric for team prediction:** $xG+$
- **Best aggregation method:** at-least-one-per-possession
- **Key insight:** Shot creation is more stable than finishing
- **Why this matters:** match analysis, player evaluation, recruitment, and forecasting
- **Future work:**

Conclusions

- **Best metric for team prediction:** $xG+$
- **Best aggregation method:** at-least-one-per-possession
- **Key insight:** Shot creation is more stable than finishing
- **Why this matters:** match analysis, player evaluation, recruitment, and forecasting
- **Future work:**
 - Sequence models for xS (point processes / hazard models)

Conclusions

- **Best metric for team prediction:** $xG+$
- **Best aggregation method:** at-least-one-per-possession
- **Key insight:** Shot creation is more stable than finishing
- **Why this matters:** match analysis, player evaluation, recruitment, and forecasting
- **Future work:**
 - Sequence models for xS (point processes / hazard models)
 - Hierarchical team/player effects to control for selection bias

Conclusions

- **Best metric for team prediction:** $xG+$
- **Best aggregation method:** at-least-one-per-possession
- **Key insight:** Shot creation is more stable than finishing
- **Why this matters:** match analysis, player evaluation, recruitment, and forecasting
- **Future work:**
 - Sequence models for xS (point processes / hazard models)
 - Hierarchical team/player effects to control for selection bias
 - Value player actions (shoot vs pass vs carry) with counterfactual $xG+$

Conclusions

- **Best metric for team prediction:** $xG+$
- **Best aggregation method:** at-least-one-per-possession
- **Key insight:** Shot creation is more stable than finishing
- **Why this matters:** match analysis, player evaluation, recruitment, and forecasting
- **Future work:**
 - Sequence models for xS (point processes / hazard models)
 - Hierarchical team/player effects to control for selection bias
 - Value player actions (shoot vs pass vs carry) with counterfactual $xG+$
 - Validate in other leagues and deploy in real time

Questions?



Link to paper, code, and slides

Acknowledgements

- Special thanks to our data providers at Gradient Sports
- All work supported by the Wharton Sports Analytics & Business Initiative (WSABI)

Supplementary Material

Mixed Effects Modeling (Per Fold)

Fitted on training data for each fold:

$$\text{metric} \sim (1|\text{season}) + (1|\text{season:team}) + (1|\text{season:opp}) + \text{home}$$

Extracted effects:

- Team attack (per season)
- Opponent defense (per season)
- Season effect
- Home field advantage

Secondary Poisson Model

Train Poisson regression on adjusted metrics:

$$\text{goals} \sim \text{home} + \text{season} + \text{team_off} + \text{opp_def}$$

Purpose: Assess predictive utility of each adjusted metric on actual goals

Table: Mean Absolute Error (MAE) by Metric and Aggregation Method

Aggregation Method	xG+	xS	xG
At-least-one-per-possession	1.86	1.87	1.89
Max-per-possession	1.86	1.86	1.89
Sum-of-shots			1.87

90% Squared Error Intervals

Table: 90% squared error intervals by metric and aggregation method

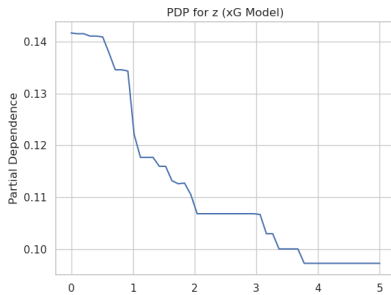
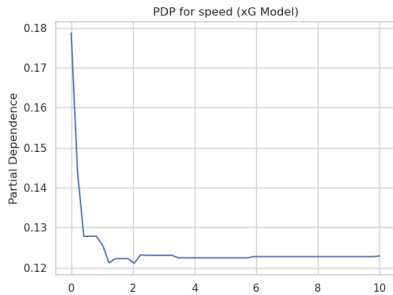
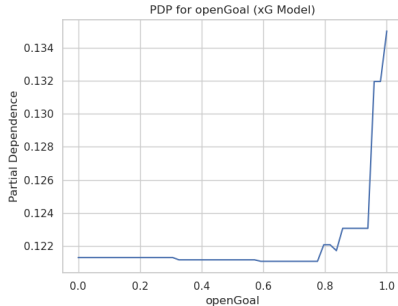
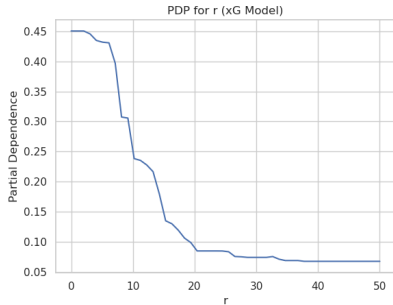
Aggregation Method	xG+	xS	xG
At-least-one-per-possession	(0.800, 2.25)	(0.823, 2.28)	(0.819, 2.32)
Max-per-possession	(0.792, 2.26)	(0.817, 2.27)	(0.802, 2.31)
Sum-of-shots			(0.822, 2.30)

Training-Sample Robustness

Table: Min-max intervals for MSE difference vs sum-of-xG

Aggregation Method	xG+ diff	xS diff	xG diff
At-least-one-per-possession	(-0.112, -0.061)	(-0.122, -0.072)	(-0.146, -0.096)
Max-per-possession	(-0.093, -0.046)	(-0.108, -0.057)	(-0.106, -0.050)

xG Partial Dependence Plots



xS Partial Dependence Plots

